

ALLEGRA FARRAR

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Allegra is a researcher passionate about reimagining space technology to address the uncertainties of Earth's climate. Through developing real-time computational and mission design strategies for Earth-observing satellites, she aims to improve the study of tropical cyclones and transient, unpredictable climate trends.

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

Ph.D. Candidate, Aeronautics and Astronautics

Expected May 2027

Space Enabled Research Group; Advisor: Danielle Wood

Master of Science in Aeronautics and Astronautics

May 2022

Astrodynamics, Space Robotics, and Controls Laboratory; Advisor: Richard Linares

The George Washington University, Washington, DC

Bachelor of Science, Mechanical Engineering (*Summa Cum Laude*)

May 2020

RESEARCH PROJECTS

Adaptive Sensing for Earth Science Satellites

NASA Jet Propulsion Laboratory (JPL), Grant Project

2025-Present

- Designs a Bayesian inference-driven adaptive sampling model that optimizes satellite observing strategy based on learned relationships as data is collected, prioritizing regional uncertainty over pre-programmed selection of known phenomena
- Methodology directs sensing assets using real-time observation and analysis to target areas of developing tropical cyclones (TCs) requiring higher resolution or signal-to-noise measurements. Extends directed and localized in-situ approaches (e.g., aircraft, buoy) proven effective for TC and planetary boundary layer studies to real-time, spaceborne context.
- In coordination with the Caribbean Coastal Ocean Observing System (CARICOOS), an active regional decision-making entity in extreme atmospheric weather and coastal management, to quantify the connection between satellite measurements and their impact on real-time decision capability for preparation and warning systems.

Remote Sensing Data Assessment for Tropical Cyclone Prediction in the Caribbean Islands

Space Enabled Research Group, Research Assistant Project

2024-Present

- Leading investigation into missed sensing opportunities in CYGNSS (Cyclone Global Navigation Satellite System) data by adaptively modifying onboard quality flag criteria, recovering observations currently discarded under strict flagging rules and increasing usable data coverage for storm prediction
- Designing a method for real-time scientific use of Level 1 CYGNSS data processed onboard the satellite in contrast to heavily filtered Level 2 products currently relied on by forecast models while accounting for onboard power, storage, and compute constraints of the satellite platform
- Collaborating with the CYGNSS mission PI, faculty PI, and university research partners to define and implement these data processing changes for TCs and other natural hazard applications. Also working with the CARICOOS team in Puerto Rico to quantify how integrating this recovered satellite data with coastal monitoring improves storm-response decision-making before, during, and after storm events

Adaptive Sampling for Uncertain Environments,

Model-based Embedded Robotic Systems Group, Research Project

2023-2024

- Designed adaptive sampling algorithms for autonomous underwater vehicles, fusing domain knowledge with probabilistic programming models to navigate observation strategy under uncertain ocean terrain
- Engineered efficient search techniques to scale algorithm performance across large state spaces, forming a computational foundation extended in ongoing PhD thesis work

Iterative LQR Method for Planetary Entry Uncertainty

Astrodynamics, Robotics, and Controls Lab, Masters Thesis

2020-2022

- Established a closed-loop trajectory optimizer capable of incorporating nonlinear atmospheric models and navigational data uncertainty to demonstrate a reliable, robust, and computationally reasonable method with certifiable guarantees for a safe Mars landing of a crewed vehicle.
- Applied an iterative linear quadratic regulator for closed-loop trajectory optimization, incorporating target uncertain and extreme conditions (large-mass vehicles, uncertain atmospheric and weather conditions, etc.) for autonomous control during entry.
- Several flight simulations revealed reliable performance with guaranteed uncertainty bounds.

PUBLICATIONS & PRESENTATIONS

- Farrar, A.,** Ruf, C., Balasubramaniam, R., Posselt, D., Kruczkiewicz, A., Wood, D. (2025). *Towards Adaptive Sensing: Identifying High-Utility Observations from GNSS-R Satellites using Bayesian Inference* (American Geophysical Union 2025). (Poster)
- Farrar, A.,** Wood, D. (2025). *An Onboard Analysis Model for Adaptive Remote Sensing of Tropical Cyclones* (76th International Astronautical Congress (IAC): Sydney, Australia). (Paper & Presentation)
- Farrar, A.,** Williams B. (2024). *Closing the Gap: Applying a Causal Model to Address Data Sparsity of Tropical Cyclones in the Global South* (American Geophysical Union 2024). (Poster)
- Farrar, A.,** Williams, B. (2024). *Discovering Causal Models of Cyclogenesis through an Informed Structural Learning Approach* (75th International Astronautical Congress (IAC): Milan, Italy). (Paper & Poster)
- Farrar, A.** (2024). *Structural Learning for Dynamic, Sparse Data Environments* (Cognitive Robotics, Dept. of Aeronautics and Astronautics, MIT). (Lecture)
- Farrar, A.,** Wood, D. (2023). *Extreme Weather and the Community: Exploring the disproportionate impacts of tropical storms and their correlation to community resilience* (Space Enabled Designs for Justice and Development, Dept of Aeronautics and Astronautics, MIT). (Proposal & Presentation)
- Farrar, A.,** Doerr, B., Linares, R. (2022). *An iterative LQR Method for Addressing Model Uncertainty for Planetary Entry* (73rd International Astronautical Congress (IAC): Paris, France). (Paper & Presentation)
- Farrar, A.** (2022). *An Iterative LQR Method for Addressing Model Uncertainty in the Mars Entry Problem* (Master Thesis, Dept. of Aeronautics and Astronautics, MIT).

GRANTS & PROPOSALS

Awarded

Derek Posselt (JPL PI), Danielle Wood (Co-I), Allegra Farrar (Co-I / Student Lead). "An Adaptive Sensing Framework for Improved Observing and Understanding of Tropical Cyclones." NASA JPL Strategic University Research Partnership Grant, 2025—2026. \$60,000.

Not Funded

Allegra Farrar (FI), Brian Williams (PI), Sai Ravela (Co-I). "Learning Causal Models of Cyclogenesis." NASA FINESST submitted 2024. Not funded.

AWARDS & FELLOWSHIPS

Graduate

NSF Graduate Research Fellow (2020) | MIT Presidential Fellow (2020) | GEM Fellow (2020) | Matthew Isakowitz Fellow (2022) | AeroAstro Outstanding Leadership Award (2023) | Sigma Xi (2024)

Undergraduate

Christine Darden Scholar, <https://vimeo.com/330191821> (2019) | Undergraduate Space Research Association, Distinguished Undergraduate Award (2020)

TEACHING & LEADERSHIP ROLES

Global Classroom (Angola), Teaching Assistant

January 2026

- Assisted my advisor in teaching a classroom of ~50 students from ISPTEC in Luanda, Angola on the use cases of satellite technology in developing local sustainability solution.
- Mentored subgroup of students investigating water security in heavy drought southern regions (e.g., Cunene). Using my thesis methodologies of designing real-time, local data-to-information networks they developed a project mapping aquifers for sustainable supply of clean water. I supported by teaching strategies on looking up and interpreting relevant satellite data, project design, along with presentation and pitch skills.

Attitude Determination and Control Systems, Teaching Assistant

Spring 2025

- Generated homework assignments, exams and taught weekly recitation classes centered around spacecraft attitude determination and control. This included topics such as rigid-body dynamics, Euler and Kepler equations, and state estimation (i.e., Kalman filters)

Patti Grace Smith, Near Peer Mentor:

2022-Present

- Support selected fellows with mentorship, guiding their entry into the Aerospace field. Continue to support mentees after undergrad as they work towards the next phase of their career.

Certificate in Aerospace Innovation, MIT AeroAstro:

Spring 2022

- Participated in the inaugural cohort working with a team to design a start-up for technology to address the issue orbital debris in low-Earth orbit. We came in 3rd place for the pitch competition.